

OASIS: On-Demand Hierarchical Event Memory for Streaming Video Reasoning

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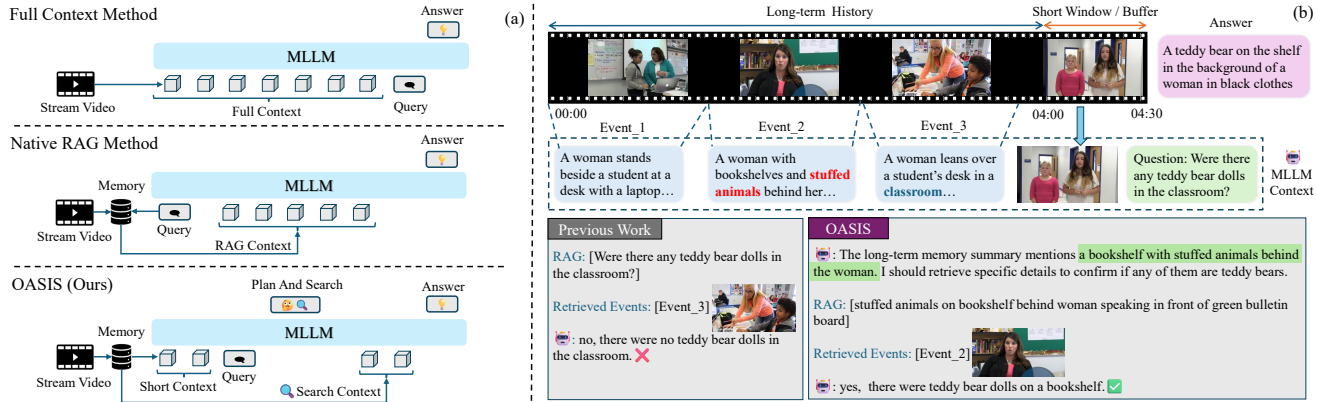


Figure 1. (a) Comparison of the OASIS framework with conventional streaming video understanding methods. **Full Context Method:** Blindly stacks all historical frames into the MLLM, leading to high cost and attention collapse. **Naive RAG Method:** Uses a fixed-policy (e.g., top-k) retrieval that is not task-adaptive. The rigid memory access policy retrieve and stack evidence into a single context window alongside the present-moment frames all the time, inevitably contaminates the reasoning process. **OASIS:** The model first performs a Coarse Reasoning using a Short Context. If insufficient, it initiates a Fine Reasoning phase, infers what additional evidence would make the answer decidable, and uses this semantic hypothesis to route a precise jump into the hierarchical event memory. (b) On the query “Were there any teddy bear dolls in the classroom?”, Naive RAG embeds the raw query and broadly matches “in classroom”, retrieving semantically irrelevant memories. OASIS infer what evidence is needed to solve the problem based on memory, yielding a precise answer.

Abstract

Streaming video reasoning requires models to operate in a setting where history grows without bound while meaningful evidence remains scarce. In such a landscape, relevant signal is like an oasis – small, critical, and easily lost in a desert of redundancy. Enlarging memory only widens the desert; aggressive compression dries up the oasis. The real difficulty lies in discovering where to look, not how much to remember. We therefore introduce OASIS, a novel framework for streaming video reasoning that tackles this challenge through structured, on-demand retrieval. It organizes streaming history into hierarchical events and performs reasoning as controlled refinement – short-context

inference first, followed by semantically grounded retrieval only when uncertainty arises. As the retrieval is driven by high-level intent rather than embedding similarity, the retrieved memory is substantially more accurate and less noisy. Additionally, the mechanism is plug-and-play, training-free, and readily attaches to different streaming MLLM backbones. Experiments across multiple benchmarks and backbones show that OASIS achieves strong gains in long-horizon accuracy and compositional reasoning with bounded token cost and low request delay. Code is available at <https://github.com/Solus-sano/OASIS>.

1. Introduction

Streaming video reasoning lies in the native regime of temporal intelligence – the world does not present itself in shuffled clips, but as a single continuous stream. Autonomous driving, security operators, AR glasses, and embodied agents: all

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of them need to interpret the world as it unfolds continuously, with no ability to scroll back. This regime is fundamentally hard as the temporal context grows without bound while the information useful to each query is extremely sparse. On one hand, naïvely retaining all past frames causes attention to drown in a desert of noises. On the other hand, compressing them permanently could erase subtle but decisive evidence. Simply scaling the token budget or the compression rate only shifts where the failure happens, without resolving the core difficulty: long streaming video is mostly redundant, yet each query usually depends on some tiny and highly localized regions in the past.

Unlike machines, humans do not attempt to retain the totality of past perception. Instead, we let almost all of it recede into latent long-term memory and remain focused on the immediate slice of the world that we are currently processing. Specifically, we operate on a short working window first, and only when we become uncertain do we actively recall a *specific* prior moment – often a small patch in the timeline that contains the decisive fact. In other words, humans do not carry the whole desert; we *find an oasis* when the present demands it. Therefore, the critical operation in streaming video reasoning is not storage, compression, or token scheduling, but *on-demand identification* of the correct region in long-term experience.

This observation motivates a different computational primitive. The essential capability in streaming video reasoning is not to store more history, but to *pinpoint the decisive regions in it*. The proper paradigm is thus on-demand and hierarchical event-level retrieval, where the short context serves as the operational substrate, and the long history is accessed only when the present underdetermines the answer. Viewed this way, streaming video reasoning is best understood as a *temporal routing* problem – identifying the unique prior event that settles the query.

We materialize this principle by proposing OASIS – **On-Demand Architected Semantic Indexed Slices** – a hierarchical event memory that makes “oasis seeking” explicit. OASIS maintains three complementary tiers of temporal representation: (i) a high-fidelity *short* window that captures the recent present, (ii) a *medium*-resolution buffer that preserves recent but non-immediate structure, and (iii) a multi-resolution event hierarchy that summarizes *long-term* history. As the stream unfolds, this hierarchy is updated online via segmentation and structural merging, producing a compact but semantically navigable memory of the past.

Critically, OASIS executes reasoning in two phases, as shown in Figure 1(a). The system begins with a short working context, *i.e.* the coarse reasoning stage, where it answers using only the immediately accessible recent window and long-term summary. As shown in Figure 2, this preserves focus, limits attention noises and attains the fidelity of the present. When the short context fails to provide sufficient

signals, OASIS enters a fine reasoning phase. Rather than inflating the window wholesale, it infers what additional evidence would make the answer decidable, and uses this semantic hypothesis to route a precise jump into the hierarchical event memory. The retrieved evidence is thus not obtained through brute-force similarity search, but via semantically guided reasoning that targets the most relevant historical region. As illustrated in Figure 1(b), unlike naïve RAG-based retrieval that often fetches visually similar but semantically irrelevant memories and yields wrong answers, OASIS leverages subtle yet decisive cues to retrieve the correct past event, leading to accurate reasoning over long, cluttered streams. This disciplined refinement neutralized both dominant failure modes of streaming video models: attention collapse from unbounded windows, and semantic misalignment from similarity-based retrieval.

In addition, our formulation is plug-and-play and training-free. OASIS does not alter the underlying multimodal large model (MLLM), nor does it require any additional training: it sits on top as a structural memory policy, and can be paired with different streaming MLLM backbones as-is. Our empirical results demonstrate that OASIS consistently improves long-horizon correctness, lifts reasoning accuracy on compositional queries, and simultaneously reduces the active window budgets by a large margin. These results validate that structured access is the key determinant of scaling temporal competence. In summary, our contributions are threefold:

- We establish a fundamental reframing of streaming video reasoning as a temporal routing task – the central difficulty is not capacity, but surfacing the correct region in a vast history.
- We introduce OASIS, which instantiates this view via a hierarchical event memory and a two-stage refinement protocol that achieves semantically guided, on-demand retrieval.
- We empirically validate OASIS across multiple MLLM backbones, showing that it remains training-free while achieving strong long-horizon gains, bounded token cost, and low request delay.

2. Related Work

2.1. Long-Video Understanding

The advent of large language models (LLMs) has catalyzed a paradigm shift across diverse domains [11, 13, 32, 46]. In video understanding, MLLMs [6, 10, 12, 18, 33, 38, 47, 48] have made significant strides, yet long videos remain challenging due to the large number of visual tokens. Existing approaches typically employ sparse frame sampling or temporal pooling [2, 4, 5, 10, 20, 21, 34, 36, 44, 45, 54] to fit within limited context windows. While effective for global summarization, these strategies often discard fine-grained

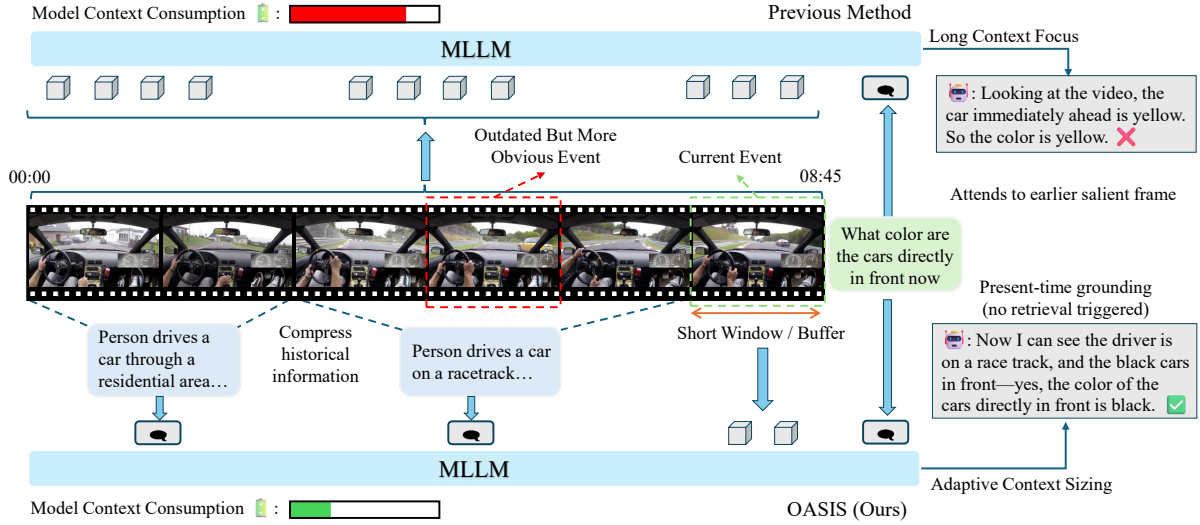


Figure 2. A critical failure case of prior methods under real-time querying. When asked, “What color are the cars directly in front now?”, the baseline model (top) processes an excessively large context, but its attention is distracted by a previously salient event (the “other car”), causing the relevant signal to be drowned in noise and yielding an incorrect answer. In contrast, OASIS (bottom) adopts an Adaptive Short Context that constrains attention to the most relevant Short Window/Memory, enabling stable grounding and correct prediction.

temporal evidence critical for downstream reasoning. Recent works instead explore query-adaptive retrieval over full videos. VideoTree [35] organizes videos into coarse-to-fine trees to preferentially allocate computation to query-relevant regions. VideoMind [19] further combines agentic planning with temporal grounding to localize relevant segments in pre-recorded videos. However, both methods assume fixed, fully observed videos and require reprocessing when the stream grows. In contrast, OASIS maintains an online, bounded Event Forest that supports cross-query reuse under continuous streaming without rebuilding search structures.

2.2. Streaming Video Understanding

Streaming video understanding requires real-time responsiveness and online interactivity. Early approaches [3, 14, 41, 42, 56] attempt to accommodate all history via stacking or compression, yet escalating token budgets force aggressive summarization, incurring irreversible loss of fine-grained details. Alternative methods [25, 40, 49, 50, 57] employ external memory with hierarchical structures, but their rigid retrieval-and-stack policy invariably concatenates retrieved cues with current frames, entangling heterogeneous temporal information and inducing attention collapse during reasoning.

2.3. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) enriches LLM responses by retrieving external knowledge and supplying it as conditioning context [7, 9, 20, 22, 28, 39]. However, standard RAG critically assumes that the user’s raw query

is already an effective retrieval query—an assumption that breaks down in complex, multi-step interactions. To mitigate this issue, Iterative RAG [29] enables the model to propose an initial hypothesis, then repeatedly retrieve new evidence to refine the answer until reaching self-consistency. Inspired by iterative and agentic retrieval paradigms, OASIS instantiates intent-driven retrieval for streaming video reasoning with an online-maintained hierarchical memory.

3. Method

In this paper, we propose OASIS, which is a training-free agent system capable of processing streaming video and answering the interleaved questions raised on any moment of the video. Specifically, a hierarchical dynamic memory is introduced to summarize the video stream into a compact yet semantically navigable representation of the past. This memory includes a high-fidelity short window, a medium-resolution buffer, a multi-resolution event hierarchy and a QA summary to capture different resolutions of memory. Based on the memory, OASIS introduces a two-phase reasoning policy which first performs coarse reasoning on the recent window and summative memory and a fine reasoning process to retrieve evidence in the multi-resolution event hierarchy. The overall framework of OASIS is shown in Figure 3.

3.1. Hierarchical Event Memory

To address the attention collapse problem in multi-modal large language models (MLLM) and achieve efficient long-term memory management, OASIS adopts a hierarchical

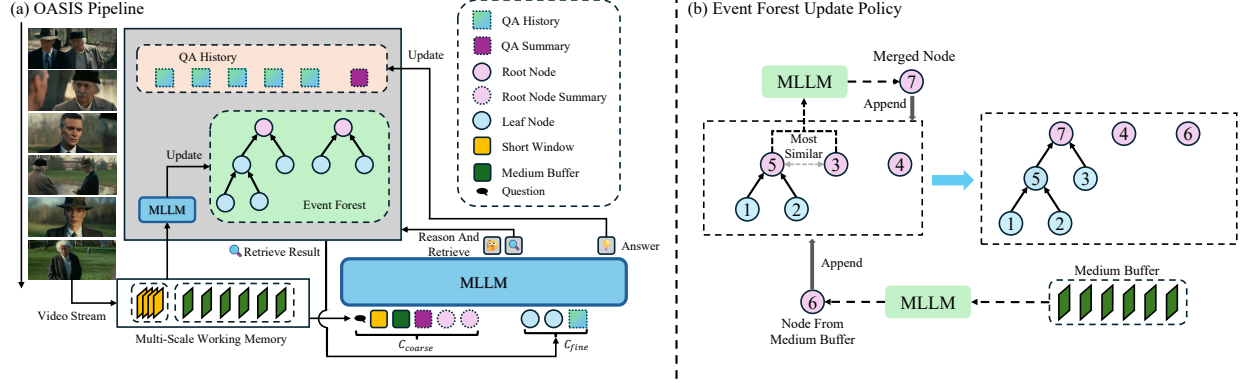


Figure 3. (a) An overview of OASIS’s multi-resolution event hierarchy and the two-phase reasoning policy that executes adaptive temporal routing. (b) The dynamic merging strategy used for the online and bounded maintenance of the event forest.

event memory structure, including a high-fidelity short window, a medium-resolution buffer, and a multi-resolution event hierarchy to capture the information in video frames.

Short window. Given a video $\mathbf{V} = \{\mathbf{X}_t\}_{t=1}^n$ consisting of n video frames (noted as $\mathbf{X}_t$) and the i -th question raised on the t_i -th frame, we introduce a high-fidelity short window to manage the current frames in streaming video. The short window retains the current frames $\mathbf{W}_s = \{\mathbf{X}_t\}_{t=t_i-\tau_s}^{t_i}$ on the t_i -th frame with a frame rate r_s , where τ_s is a small number indicating the interval of current frames. The short window preserves the fine-grained visual and temporal details required for accurate, present-moment reasoning.

Medium-resolution Buffer. Furthermore, OASIS introduces a medium-resolution buffer to preserve recent but non-immediate structure. This buffer retains the frames from $\mathbf{W}_m = \{\mathbf{X}_t\}_{t=t_i-\tau_m}^{t_i}$ with a smaller frame rate r_m ($r_m < r_s$), where τ_m denotes the interval of medium-resolution frames. In contrast to the short window, the medium-resolution buffer exploits information from a wider range of frames before the current frame, enlarging the perceptual field of MLLM.

Multi-resolution event hierarchy. Employing the short-term window and the medium-resolution buffer allow the MLLM to maintain context for recent events. However, for streaming videos, this naive accumulation of information is unsustainable. As the video progresses, the memory required grows linearly, quickly surpassing the context window limitations of modern MLLMs and leading to a significant scalability bottleneck. To overcome this, we introduce a multi-resolution event hierarchy, organized as an Event Forest, which summarizes and structures events to maintain a compact yet comprehensive history.

The Event Forest $\mathcal{F}$ consists of a series of event nodes, which are populated on-the-fly by processing the video stream in sequential, non-overlapping temporal windows. Specifically, for every incoming window of t_m frames, we uniform sample n_f key frames and abstract its content into

a new event node, which is immediately added to the forest. This ensures that our representation stays synchronized with the live stream. Initially, the j -th node $\mathbf{R}_j$ is constructed by:

$$\mathbf{R}_j = ([t_s^{(j)}, t_e^{(j)}], \mathbf{F}_j, s_j, \mathbf{e}_j, d_j). \quad (1)$$

Here $[t_s^{(j)}, t_e^{(j)}]$ indicates the time interval of videos abstracted in the node $\mathbf{R}_j$. Note that $t_s^{(j)} - t_e^{(j)} = t_m$ initially. $\mathbf{F}_j$ is a 4D tensor stack of n_f key frames. s_j is a text summary generated by MLLM that describes the observable facts of the current window. $\mathbf{e}_j = \mathbf{E}(s_j)$ is the embedding vector of this summary output by an embedding encoder $\mathbf{E}$. d_j denotes the hierarchical level of the node. All nodes created directly from the video stream are initialized as level-0 leaf nodes, i.e., $d_j = 0$. We denote the set of roots as $\mathcal{R} = \{\mathbf{R}_j\}_{j=1}^{|\mathcal{R}|}$.

Node Merging. The unbounded growth of the root set would render the context retrieval for the MLLM computationally prohibitive. To prevent this issue, we introduce a bounded-complexity online merging mechanism to merge the node in the root set. This mechanism is triggered whenever the number of root nodes $|\mathcal{R}|$ exceeds a predefined threshold n_r . We employ a hierarchy-aware strategy that greedily merges the temporally adjacent root pair with the highest merging score. For the temporally adjacent root pair $(\mathbf{R}_j, \mathbf{R}_k)$ with $t_e^{(j)} = t_s^{(k)}$, the merging score is estimated as,

$$\text{score}(\mathbf{R}_j, \mathbf{R}_k) = \cos(\mathbf{e}_j, \mathbf{e}_k) - \lambda(d_j + d_k). \quad (2)$$

This scoring function elegantly balances two competing objectives. The cosine similarity function $\cos(\cdot)$ promotes the fusion of semantically and temporally coherent events. λ is a hyperparameter that penalizes the merging of nodes with high hierarchical levels with large d_j and d_k . This avoids over-abstractation and preserves the hierarchical granularity of memory.

The selected pair $(\mathbf{R}_j, \mathbf{R}_k)$ is then fused into a new parent

node $\mathbf{R}_l$ which is formulated as,

$$\begin{aligned}\mathbf{R}_l &= ([t_s^{(l)}, t_e^{(l)}], \mathbf{F}_l, \mathbf{s}_l, \mathbf{e}_l, d_l), \\ t_s^{(l)} &= t_s^{(j)}, \quad t_e^{(l)} = t_e^{(k)}, \\ \mathbf{F}_l &= \text{Uniform}(\mathbf{F}_j \oplus \mathbf{F}_k) \\ \mathbf{s}_l &= \text{Merge}(\mathbf{s}_j, \mathbf{s}_k), \\ \mathbf{e}_l &= \mathbf{E}(\mathbf{s}_l), \\ d_l &= \max(d_j, d_k) + 1.\end{aligned}\quad (3)$$

Here $\oplus$ is the concatenation operation. $\text{Uniform}(\cdot)$ is the uniform frame sampling operation. $\text{Merge}(\cdot)$ is the summary merge process performed by LLMs. Subsequently, $\mathbf{R}_j$ and $\mathbf{R}_k$ are removed from the root set, while the new node $\mathbf{R}_l$ is added to the root set. The parent-children relation between $\mathbf{R}_l$ and $(\mathbf{R}_j, \mathbf{R}_k)$ is also updated in our event forest $\mathcal{F}$. This design allows the event forest to adaptively organize long video streams and generate a limited-size summary of the long video by aggregating the summary $\{s_j\}_{j=1}^{|\mathcal{R}|}$ of root nodes.

QA Summary. We also propose a question-answer(QA) summary to capture the historical question-answer information. For each question q_i raised on the t_i -th frame and the corresponding answer a_i , OASIS aggregates the content of the question and frame into the current QA Summary by an LLM,

$$s_{\text{qa}} \leftarrow \text{LLM}(s_{\text{qa}}, (q_i, a_i)). \quad (4)$$

The QA summary serves to provide a low-cost dialogue context for MLLM and ensure naming consistency in multi-turn dialogues. Additionally, we also maintain all QA history $\mathcal{Q} = \{q_j, a_j\}_{j=1}^i$. Specifically, we estimate an embedding for each question-answer pair by $\mathbf{e}_j^q = \mathbf{E}(q_j \oplus a_j)$ for subsequent retrieval.

3.2. Two-phase Reasoning

To achieve the ‘‘oasis seeking’’ explicit in memory context, OASIS proposes a two-phase reasoning policy, which operates in two phases. OASIS first performs a coarse reasoning on immediately accessible memory and output whether the information is sufficient to answer the question. If so, OASIS directly outputs the answer. Otherwise, it will enter the fine Reasoning stage, which retrieves long-term evidence in the multi-resolution event hierarchy to support the answer.

Coarse Reasoning. When a question q_i is raised, OASIS applies an MLLM to generate the model response using only the immediately accessible memory. Concretely, given the immediately accessible memory $\mathbf{C}_{\text{coarse}}$ (comprising all frames in short window $\mathbf{W}_s$ and medium-resolution buffer $\mathbf{W}_m$, summaries of all root nodes $\{s_j\}_{j=1}^{|\mathcal{R}|}$ and the QA history summary s_{qa}) and the raised question q_i , the MLLM first outputs a preliminary response with a coarse reasoning process. This process is formulated as,

$$\text{Think}, \hat{a}_i \leftarrow \text{MLLM}(p_{\text{sys}}, \mathbf{C}_{\text{coarse}}, q_i), \quad (5)$$

where p_{sys} is the system prompt. Benefits from our event forest structure, the size of $\{s_j\}_{j=1}^{|\mathcal{R}|}$ is limited, which avoids prohibitive token overhead and attention collapse caused by long input contexts. Subsequently, we modify p to prompt the model to consider two cases. If the immediately accessible memory (e.g., current frames or general knowledge) contains all the facts needed to answer q_i , the model is prompted to directly generate the final answer. If the information is insufficient or the question pertains to specific details from the distant past, the model is instructed not to guess. Instead, it must signal the need for a deeper memory search by invoking a specific tool. The discrimination between these two cases is achieved by parsing special tokens in the model’s initial response $\hat{a}$. A response wrapped in $\langle \text{answer} \rangle \dots \langle \text{/answer} \rangle$ tags signals that the model has confidently formulated a direct answer, which is then finalized. In contrast, A response containing the $\langle \text{tool_call} \rangle$ token indicates that the model has identified the information as insufficient, thereby triggering the fine reasoning phase. This design forces the model to prioritize high-resolution, recent information, fundamentally suppressing the attentional misalignment of answering the current question with old evidence.

Fine Reasoning. The fine reasoning is processed in the planning, retrieving, and answering manner. OASIS first generates a retrieval query I_i , which can be directly phased from the initial response of the model $\hat{a}_i$. The retrieval query I_i is a task-oriented, high-level semantic internal query, providing more precise retrieval clues than the original question q_i . (e.g., for q_i ‘‘Was there anyone there just now?’’ and the video frames of the living room scene, the I_i is set to ‘‘people in the living room’’). Subsequently, we encode I_i with the embedding encoder $\mathbf{E}$ to the embedding $\mathbf{E}(I_i)$. This embedding is used to estimate the retrieval score to select the most relevant content in the multi-resolution event hierarchy and the QA history.

For the multi-resolution event hierarchy, we select the k_f nodes with I_i from the event forest $\mathcal{F}$. To ensure informational diversity and avoid retrieving nodes with direct lineage (e.g., a parent and its child), we employ a greedy pruning strategy. We iteratively select the node with the highest similarity score, $\arg\max_j \cos(\mathbf{E}(I_i), \mathbf{e}_j)$, and then remove its ancestors and descendants from the candidate pool for subsequent selections. Concurrently, for the QA history $\mathcal{Q}$, we retrieve the k_q historical question-answer pairs with top k_q retrieval scores, $\text{TopK}_j \cos(\mathbf{E}(I_i), \mathbf{e}_j^q)$, with I_i .

The retrieval content $\mathbf{C}_{\text{fine}}$ includes the key frames tensor of the k_f selected nodes and the retrieved k_q QA pairs. Obtained the retrieved content, we update the input context for MLLM and generate the final answer a_i ,

$$a_i \leftarrow \text{MLLM}(p_{\text{sys}}, \mathbf{C}_{\text{coarse}}, q_i, \mathbf{C}_{\text{fine}}). \quad (6)$$

This strategic retrieval is the key to achieving accurate and

temporally consistent understanding in dynamic video environments.

4. Experiment

4.1. Benchmark

To comprehensively evaluate OASIS’s ability, we select three streaming video benchmarks: OVO-Bench [23], StreamingBench [16], and StreamBench [40].

OVO-Bench. OVO-Bench focuses on time-sensitive online understanding. We focus on the Real-time Understanding and Backward Tracing scenarios. Real-Time Understanding directly probes the model’s ability to overcome attention collapse and precisely ground the present moment. Backward Tracing directly tests its long-term memory retrieval.

StreamingBench. StreamingBench systematically evaluates streaming understanding across real-time vision, multi-source input, and context comprehension. It simulates continuous multi-turn QA interactions by providing 5 questions per video. We adhere to the official All Context setting, where the entire video stream is made available to the model.

StreamBench. We additionally evaluate on StreamBench, a open-ended benchmark; detailed settings and results are deferred to the supplementary material due to space limits.

4.2. Implementation Details

Our experiments are based on the Qwen3-VL-8B [27], Qwen2.5-VL-7B [1], and GLM-4.6V [8]. For OASIS’s hierarchical memory, we set the short window ($\mathbf{W}_s$) duration to $\tau_s = 8s$ at $r_s = 2$ fps, and the medium-resolution buffer ($\mathbf{W}_m$) duration to $\tau_m = 32s$ at $r_m = 1$ fps. Each event node in the Event Forest ($\mathcal{F}$) has a keyframe budget of $n_f = 16$, the root set is capped at $n_r = 4$ nodes, and the merge penalty is $\lambda = 0.1$. To maximize efficiency, the same MLLM instance is used for all management tasks (summarization, merging, QA updates) and final reasoning. We use Qwen3-Embedding-0.6B [55] as the embedding encoder $\mathbf{E}$. During the Fine Reasoning stage, we retrieve $k_f = 2$ event nodes and $k_q = 1$ QA pair. All experiments were conducted on a single NVIDIA A800 80GB GPU.

4.3. Baselines

To provide a comprehensive comparison, we benchmark OASIS against two categories of state-of-the-art (SOTA) models. First, we compare against SOTA Offline MLLMs to establish a performance ceiling based on general visual reasoning. This includes top-tier open-source models: LongVA [52], LLaMA-VID [15], LLaVA-Hound [53], LLaVA-Video-7B [54], InternVL2 [4], LLaVA-OneVision-7B [10], VideoLLaMA2 [48], MiniCPM-V 2.6 [43], VILA-1.5 [17], InternLM-XCP2.5 [51], MovieChat [30], FreeVA [37], Qwen2.5-VL [1], Qwen2-VL [33]. As well as powerful closed-source models like Gemini-1.5 Pro [31] and GPT-

4o [24]. Second, we also conduct a head-to-head comparison with SOTA Online MLLMs. This includes FlashVStream [49], VideoLLM-online [3], Dispider [26], and StreamChat [40]. We strictly adhere to the online protocol to simulate online inference: for a query q_i at timestamp t_i , the model is only given access to the video segment $\mathbf{V}_{t_i} = \{\mathbf{X}_t\}_{t=1}^{t_i}$ and the preceding QA history $\mathcal{Q}_{t_i}$.

4.4. Main Results

Results on OVO-Bench. We compare OASIS against existing baselines on OVO-Bench in Table 1. When equipped with the OASIS framework, both base models (Qwen2.5-VL-7B and Qwen3-VL-8B) achieve improvements in the Perception and Backward scenarios. The most compelling result comes from the Perception scenario, which explicitly tests the model’s ability to ground the present moment without being distracted by history—a direct test of the attention collapse dilemma. With OASIS, Qwen3-VL-8B achieves a massive +11.35 gain. This result strongly proves that a superior framework can outperform brute-force context OASIS’s Coarse Reasoning stage, by forcing a focus on the high-resolution short Window, which fundamentally solves the attention collapse problem. Concurrently, in the Backward scenario, OASIS delivers a +6.02 improvement, demonstrating that its Fine Reasoning stage with plan-driven RAG effectively overcomes the irreversible information loss problem. We note that the gains on Qwen2.5-VL-7B are more modest. As OASIS’s agentic policy heavily relies on the base LLM’s instruction-following capabilities, which are stronger in the Qwen3-VL model.

Cross-backbone generalization. To verify that OASIS is not tied to the Qwen family, we further evaluate GLM-4.6V on OVO-Bench in Table 3. OASIS improves GLM-4.6V by +13.43 on Perception and +3.92 on Backward, showing that the gains come from the proposed memory policy rather than a backbone-specific implementation. Together with the Qwen results above, this supports the plug-and-play nature of OASIS across different MLLM families.

Results on StreamingBench. We also evaluate on StreamingBench in Table 2. OASIS achieves improvements across all four sub-tasks. The gains are particularly pronounced in the MCU task. This task is specifically designed to induce attention collapse by presenting highly similar, continuous footage. OASIS’s success here is again attributable to our two-stage policy. Furthermore, the improvements in ACU task demonstrate that OASIS’s short window is effective at capturing real-time, transient events. The significant gains in Sequential Question Answering (SQA) prove that OASIS’s Plan-driven RAG and QA Summary mechanism are superior to naïve RAG for handling co-reference and long-term dependencies in multi-turn dialogues.

Results on StreamBench. We also observe consistent gains on the more open-ended StreamBench benchmark;

Table 1. Comparison on the OVO-Bench benchmark. We report the accuracy(%) on the Perception and Backward subset. The best results are indicated by **bolding**. ‡ denotes results reproduced by us, while others are taken from prior works.

Model	frames	Perception							Backward			
		OCR	ACR	ATR	STU	FPD	OJR	Avg	EPM	ASI	HLD	Avg
Gemini 1.5 Pro	1fps	85.91	66.97	79.31	58.43	63.37	61.96	69.32	58.59	76.35	52.64	62.54
GPT-4o	64	69.80	64.22	71.55	51.12	70.30	59.78	64.46	57.91	75.68	48.66	60.75
LLaVA-Video-7B	64	69.13	58.72	68.83	49.44	74.26	59.78	62.32	56.23	53.47	51.53	53.74
LLaVA-OneVision-7B	64	66.44	57.80	73.28	53.37	71.29	61.96	62.04	56.24	55.41	21.51	43.71
Qwen2-VL-7B	64	40.40	50.46	63.76	47.19	66.34	53.55	50.98	48.71	35.45	45.03	43.73
Qwen2-VL-72B	64	65.77	60.55	69.83	51.69	69.31	54.35	61.92	52.53	60.81	57.53	56.95
Flash-VStream-7B	1fps	24.16	29.36	28.45	33.71	25.74	28.80	28.37	39.06	37.16	5.91	27.38
VideoLLM-online-8B	2fps	8.05	2.83	12.04	14.04	45.54	21.10	22.22	18.80	12.18	17.73	16.24
StreamChat [‡]	15fps	51.68	47.71	58.62	41.01	61.06	47.83	50.12	50.17	54.05	37.63	47.39
Dispider	1fps	57.72	49.54	62.07	44.94	61.39	51.63	54.55	48.48	55.41	45.43	36.06
Qwen2.5-VL-7B [‡]	0.5fps	76.51	57.80	68.10	46.63	66.34	56.52	60.93	49.83	58.11	44.62	50.24
+ OASIS	-	85.23	72.48	66.38	52.25	67.33	64.67	67.26	51.85	58.78	48.92	52.61
Qwen3-VL-8B [‡]	0.5fps	83.89	58.72	70.69	56.18	69.31	64.13	66.79	51.18	60.81	43.55	51.19
+ OASIS	-	91.95	80.73	81.03	67.42	67.33	79.89	78.14	61.95	60.14	47.31	57.21

Table 2. Comparison on StreamingBench. We report accuracy(%) on All Real-Time Video Understanding (Real-Time All), Misleading Context Understanding (MCU), Anomaly Context Understanding (ACU), and Sequential Question Answering (SQA). † denotes a dynamic frame rate: 1 fps for durations under 5 min, 0.5 fps for 5–10 min, and 0.2 fps for over 10 min. The best scores are highlighted in **bold**. ‡ denotes results reproduced by us, while others are taken from prior works.

Model	frames	Real-Time All	ACU	MCU	SQA
Gemini 1.5 pro	1 fps	75.69	51.41	40.73	54.80
GPT-4o	64	73.28	41.20	38.40	32.80
LLaVA-OneVision-7B	32	71.12	35.60	36.00	27.27
Qwen2-VL-7B	0.2–1 fps [†]	69.04	31.20	26.00	39.60
InternVL-V2-8B	16	63.72	32.00	31.20	32.32
Video-LLama2-7B	32	49.52	24.80	26.80	18.67
StreamChat [‡]	15 fps	60.14	28.80	22.40	32.00
Qwen2.5-VL-7B [‡]	0.5 fps	65.99	33.20	30.12	34.00
+ OASIS	-	70.59	38.80	36.00	46.00
Qwen3-VL-8B [‡]	0.5 fps	72.83	35.63	35.74	43.90
+ OASIS	-	78.22	42.74	49.60	48.40

Table 3. Cross-backbone generalization on OVO-Bench. We report average accuracy(%) on the Perception and Backward subsets. ‡ denotes results reproduced by us.

Model	frames	Perception Avg	Backward Avg
GLM-4.6V [‡]	0.5 fps	54.96	51.35
+ OASIS	-	68.39	55.27

detailed per-subset results and analysis are provided in the supplementary material.

Table 4. Ablation study on OVO-Bench. We quantify the impact of each memory component in our OASIS system. ‘Perception Avg’ and ‘Backward Avg’ denote the average performance on the Perception and Backward subset, respectively.

Medium Buffer	Event Forest	Short Window	Perception Avg	Backward Avg
✓	✗	✗	74.31	54.00
✗	✗	✓	77.66	51.93
✗	✓	✓	76.52	48.65
✓	✗	✓	77.18	54.15
✓	✓	✗	70.37	56.89
✓	✓	✓	78.14	57.21

Table 5. The ablation of key factors in the OASIS on the OVO-Bench dataset. ‘Native RAG’ denotes directly retrieving events with questions. ‘Flatten Memory’ denotes there is no limit to the number of root nodes in the event forest, allowing the root nodes to grow indefinitely.

Method	Perception Avg	Backward Avg
Flatten Memory	76.70	55.78
w/o RAG	76.19	55.22
Naïve RAG	78.01	56.13
OASIS	78.14	57.21

4.5. Ablations

We conducted a series of rigorous ablation experiments to deconstruct the OASIS framework and quantify the contribution of each core component. We used OVO-Bench as the primary ablation benchmark. All variants used Qwen3-VL-8B as the inference backbone.

The ablation of the memory system. We conduct ablation studies in Table 4 to quantify the contribution of each component in the memory system. When using only the high-frame-rate short window W_s , the Perception score reaches 77.66 while the Backward score is only 51.93. When using only the medium buffer W_m , Perception drops to 74.31 while Backward increases to 54.00. This comparison indicates that longer sampling improves the ability to answer questions that depend on long-term memory, but it can induce an attention collapse that degrades immediate-scene performance. After incorporating the Event Forest, Backward further improves to 57.21, demonstrating the Event Forest’s effectiveness at modeling long-term dependencies.

The ablation of the reasoning policy. We conduct an ablation study in Table 5 to validate the design of our reasoning strategy. When the RAG stage is removed—i.e., the model answers questions solely based on the short window W_s , medium buffer W_m , and the root summaries from the Event Forest, the performance on Backward questions drops noticeably. Replacing our Fine Reasoning process with a naïve RAG-based retrieval strategy, which directly uses the original question text for retrieval, reduces accuracy on OVO-Bench’s Backward track from 57.21 to 56.13. This demonstrates that generating retrieval cues conditioned on historical summaries allows the model to locate more relevant past events than using the question alone. When we replace the hierarchical Event Forest with a flattened memory structure—removing the root node limit n_r and allowing unbounded root node growth, both Perception and Backward accuracy decline. The primary reason is that excessive root nodes inflate the token count of the coarse summary, making retrieval less efficient and preventing effective parent–child deduplication during hierarchical reasoning.

Token Consumption. As shown in Figure 4, we compare the average token consumption of OASIS against a full-context baseline. The baseline’s token usage was positively correlated with video duration, reaching 29,517 tokens on StreamingBench (videos averaging 7–8 minutes) and proportionally high values on OVO-Bench and StreamBench (videos averaging 4–4.5 minutes). In contrast, OASIS’s token consumption remains relatively stable at around 10k across all datasets.

Latency and responsiveness. Beyond token efficiency, we also analyze wall-clock latency in Table 6. OASIS achieves a low Request Processing Delay (RPD) of 0.19s, indicating that query-time response can start almost once a user question arrives. On OVO-Bench, the end-to-end per-query latency increases from 4.14s for the single-pass baseline to 6.52s for OASIS because of the additional coarse-to-fine reasoning stage. Importantly, the retrieval itself is lightweight, taking only 0.11s; the dominant query-time cost remains MLLM inference (3.45s coarse and 3.06s fine when triggered). The memory-maintenance operations are query-

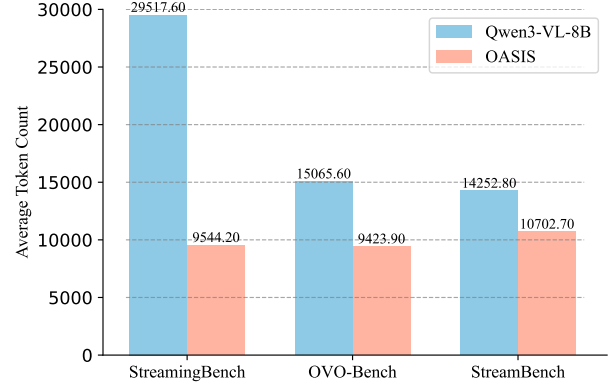


Figure 4. Comparison of Average Token Consumption per query between OASIS and the baseline. The Qwen3-VL-8B uniformly samples video at 0.5 fps.

Table 6. Latency and responsiveness of OASIS. RPD is measured on StreamBench following [40]; all other latency numbers are measured on OVO-Bench with Qwen3-VL-8B on one A800 GPU.

Metric	Time (s)
Request Processing Delay (RPD)	0.19
Baseline end-to-end per query	4.14
OASIS end-to-end per query	6.52
Memory update (generate node)	7.73
Memory update (merge root)	6.52
Embedding & RAG retrieval	0.11
MLLM inference (coarse)	3.45
MLLM inference (fine, if triggered)	3.06

independent and run asynchronously during streaming. With a 32s buffer window, the 7.73s node-generation and 6.52s root-merging costs are overlapped with incoming video.

5. Conclusion

In this work, we revisited streaming video reasoning from a structural perspective. We reformulate this task as temporal routing - the ability to locate the decisive evidence hidden within an overwhelming temporal desert. Based on this view, we proposed OASIS, a hierarchical event memory that enables selective refinement: reasoning in the short context by default, and retrieving long-term evidence only when semantically required. The model-agnostic and training-free nature of the approach allows seamless integration with existing MLLMs while delivering substantial efficiency and accuracy improvements.

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